

$$f(x) = \frac{x^3}{x^2-4} = x + \frac{4x}{x^2-4}$$

$$D_f = \mathbb{R} \setminus \{-2; 2\}$$

$$\text{Zeros: } f(x)=0 \Leftrightarrow x=0$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x^3}{x^2-4} = \frac{\infty}{\infty} = \lim_{x \rightarrow \pm\infty} \frac{x^3}{x^2(1-\frac{4}{x^2})} = \lim_{x \rightarrow \pm\infty} \frac{x}{1-\frac{4}{x^2}} = \frac{\infty}{1} = \infty$$

$$f'(x) = \left(\frac{x^3}{x^2-4} \right)' = \frac{(x^3)'(x^2-4) - (x^2-4)'x^3}{(x^2-4)^2} = \frac{3x^2(x^2-4) - 2x \cdot x^3}{(x^2-4)^2} = \frac{x^4-12x^2}{(x^2-4)^2} = \frac{x^2(x^2-12)}{(x^2-4)^2}$$

$$\text{Zeros: } f'(x) = 0 \Leftrightarrow 3x^2(x^2-12)=0 \Leftrightarrow x^2=0 \vee x^2-12 \Leftrightarrow x=0 \vee x=\pm\sqrt{12} \Leftrightarrow x=0 \vee x=\pm 3,46$$

$$f''(x) = \left(\frac{x^4-12x^2}{(x^2-4)^2} \right)' = \frac{(x^4-12x^2)'(x^2-4)^2 - (x^2-4)^2'(x^4-12x^2)}{(x^2-4)^4} = \frac{(4x^3-24x)(x^2-4)^2 - 2(x^2-4)(x^2-4)'(x^4-12x^2)}{(x^2-4)^4} =$$

$$\frac{(4x^3-24x)(x^2-4) - 2 \cdot 2x(x^4-12x^2)}{(x^2-4)^3} = \frac{4x^5-40x^3+96x-4x^5+48x^3}{(x^2-4)^3} = \frac{8x^3+96x}{(x^2-4)^3} = \frac{8x(x^2+12)}{(x^2-4)^3}$$

$$\text{Zeros: } f''(x)=0 \Leftrightarrow x=0$$

x		-3,5		-2		0		2		3,6	
f(x)	-		+	~	-	0	-	~	-	0	+
f'(x)	+ ↗	0	- ↘	~	- ↘	0	- ↘	~	- ↘	0	+ ↗
f''(x)	- ∩		- ∩	~	+ ∪	0	- ∩	~	+ ∪		+ ∪
	↗	M	↘	~	↘	Z P.I.	↘	~	↘	m	↗

